

Chapter 32

Trend Assessment of Rainfall Over Mumbai and Pune Cities



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Abstract The yearly rainfall trend for an area would be important in planning irrigation patterns, reservoir management, storm water drainage, and so on, as well as understanding the influence of climate change on rainfall in a region. Mumbai and Pune are two metropolitan cities in Maharashtra state that have developed in the last 20–30 year. The present study intends to examine the precipitation pattern in Mumbai and Pune. Monthly and yearly precipitation data values from the years 2000–2020 were utilized for the study. Basic statistical tests are used to assess the trend of rainfall or its absence, and rainfall trend analysis is done using linear regression analysis, Mann-Kendall analysis, and Sen's Slope estimator. The results of trend analysis have shown an increasing trend in the Annual rainfall, non-monsoon, southwest monsoon, and northeast monsoon in both Mumbai and Pune. In both cities, a decreasing trend has been observed during the pre-monsoon season. Reliability analysis is performed to provide statistics for drinking water scheme design (90% dependability) and irrigation requirement calculation (75% dependability).

Keywords Trend analysis · Rainfall · Hydrologic hazards

32.1 Introduction

India's agriculture and allied sectors, food security, and energy security are critically dependent on the timely availability of water and a favorable environment. Rainfall plays a significant role in satisfying needs of agriculture, residential water supply, hydroelectric power generation, and industries (Gandhre et al. 2020; John et al.

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2021a, b). Global climate change may alter long-term rainfall patterns that affect water supply, as well as the risk of increased droughts and floods (Pu et al. 2021; Singh et al. 2022; Wallwork et al. 2022). The south west (S-W) monsoon, which provides for approximately 80% of the total precipitation in the nation, is crucial for the supply of freshwater for drinking and irrigation (Jagadeesh and Agrawal 2015). Changes in climate across the Indian region, particularly the S-W monsoon, will have a huge influence on agricultural productivity, water resource management, and the country's general economy (Kumar et al. 2010; Potdar et al. 2019). Nowadays metropolitan cities have encountered severe water scarcity and declined groundwater levels a result of rapid urbanization. Several studies reported that the frequency of natural calamities such as floods and droughts was found to be increasing in the Indian region (Joshi and Pandey 2011; Mondal et al. 2015). Normal rainfall is required for appropriate agricultural productivity in order to keep food prices stable, as food prices are dependent on agricultural yields. As agriculture accounts for 16% of India's GDP, a proper monsoon is also required for economic development (Rana et al. 2013). Other economic activities like fishing and forestry are also directly affected by the amount of precipitation. India's population growth and modernization have resulted in increased water consumption. As a result, it is vital to investigate the features of these events as well as the scientific reasons behind them.

Numerous researches have been conducted to investigate the long-term pattern of rainfall over the Indian subcontinent. Mooley and Parthasarathy (1984) has studied an analysis of the All-India summer monsoon rainfall (June–September) from 1871 to 1978. Parthasarathy et al. (1994) have taken all India monthly, seasonal, and yearly homogenous data series from 1871 to 1993. They examined some data facts and long-term trends in monsoon rainfall in India. Patil et al. (2013) studied the trend of rainfall in Pune district using 47 years data from 1958 to 2004. Mann-Kendall (MK) and modified MK tests for detecting trends and Sen's slope estimator was utilized for quantifying the change. There was a noticeable alteration in the spatial and temporal rainfall pattern over the Pune region during the time of analysis. Rana et al. (2012) have done trend analysis for rainfall in Delhi and Mumbai cities of India. The results showed that a slight significant decreasing trend for long-term S-W monsoon rainfall in Mumbai and both stations' average maximum daily rainfall has decreased. Jagadeesh and Agrawal (2015) investigated trends and their magnitude using Mann-Kendall test and Sen slope methods. The overall analysis shows an increasing trend in the east and a more decreasing trend in the south and west compared to the north of the catchment from eight rain gage stations in Kerala's Bharathapuzha catchment from 1975 to 2009. Gandhre et al. (2020) conducted a precipitation trend analysis for Maharashtra's coastal districts from 1901 to 2000, which revealed an increasing trend for Sindhudurg and Mumbai metropolis. There is no discernible trend for Thane, Raigad, Ratnagiri, or Mumbai Suburban districts. Rainfall in Mumbai has increased by 6.25 mm each year. Potdar et al. (2019) conducted a Trend Analysis of Rainfall Data for Maharashtra and Goa from 1901 to 2015, and the findings show that the long-term trend in annual period and southwest monsoon season across Maharashtra as a whole, Konkan and Goa, and

Madhya Maharashtra is increasing, while it is decreasing in Marathwada and Vidarbha.

Maharashtra state has the second largest population in India and the third largest surface area. The two largest metropolitan cities in Maharashtra are Mumbai and Pune. The population of these two cities is rapidly growing as people migrate for jobs and education. Such cities have a higher population density than towns or rural regions. A significant portion of Mumbai was built through the process of land reclamation, which is the removal of water to raise the level of the land, it is highly susceptible to hydrological hazards such as flooding (Rana et al. 2012; Shankar et al. 2021; Pandey et al. 2022). This emphasizes the need of rainfall trend analysis so that stormwater drains may be designed to drain excess precipitation or groundwater from impermeable cemented constructions such as roadways, sidewalks, and parking lots. Water consumption is increasing as a result of urbanization and economic growth. This rainfall trend analysis study will aid in water management by determining how rainfall has changed in previous years in order to meet escalating water needs.

The aim of the research was to examine the rainfall trend in Mumbai and Pune cities from 2000 to 2020 and identify overall increasing or decreasing trends, as well as to learn how the basic statistical data has changed over the given time period for the annual period, non-monsoon season, pre-monsoon season, southwest monsoon season, and northeast monsoon season. Find out the fundamental statistics about rainfall such as maximum and minimum rainfall, mean, standard deviation, skewness, and kurtosis of data gathered in various years based on the monsoon seasons. In addition, the Mann-Kendall and Sen Slope methods will be used to analyze rainfall trends.

32.2 Data and Methodology

This research looks at rainfall changes in Mumbai and Pune, two major cities in the Indian state of Maharashtra, from 2000 to 2020. Precipitation data was analyzed by categorizing it into four seasons: (1) non-monsoon (January, February, March), (2) pre-monsoon (April, May), (3) southwest monsoon (June, July, August, September), and (4) northeast monsoon (October, November, and December) then basic statistical analysis of the data is done to find out the maximum and minimum rainfall, mean, standard deviation, skewness, and Kurtosis.

Trend is defined as the long-term change in the dependent variable over a lengthy period of time or as the overall movement of a series over an extended period of time (Panda and Sahu 2019). According to recent research, the nonparametric Mann-Kendall (MK) trend test is the most extensively used approach for detecting trends. Mann (1945) developed the test first, Kendall (1975) subsequently developed the test static known as the Kendall's tau statistic. It was discovered to be a great tool for trend identification in a variety of applications (Burn and Elnur 2002).

32.3 Mann–Kendall’s Test

The M-K test is a statistical nonparametric test that is commonly used for trend analysis in climatological and hydrological time series data. There are two benefits to using this test. First, it is a nonparametric test that does not need the data to be distributed regularly. Second, due to inhomogeneous time series, the test has limited sensitivity to sudden breaks. The null hypothesis H_0 , according to this test, suggests that there is no trend (the data is independent and randomly ordered). This is compared to the alternative hypothesis H_1 , which expects a trend. The M-K statistic may be calculated as follows:

$$S = \sum_{j=1}^{n-1} \sum_{k=j+1}^n \text{sign}(x_k - x_j) \quad (32.1)$$

where

$$\text{sign}(x_k - x_j) = \begin{cases} +1, & x_k - x_j > 0 \\ 0, & x_k - x_j = 0 \\ -1, & x_k - x_j < 0 \end{cases} \quad (32.2)$$

and tau is estimated as

$$\tau = \frac{2S}{N(N-1)} \quad (32.3)$$

Variance of S is calculated by the formula:

$$\text{VAR}(S) = \frac{1}{18} [n(n-1)(2n+5)] \quad (32.4)$$

where n is the number of value points in the set, the value of test statistic Z is calculated by

$$\begin{aligned} Z &= \frac{S-1}{\sqrt{\text{VAR}(S)}} \quad \text{if } S > 0 \\ &= 0, \text{ if } S = 0 \\ &= \frac{S+1}{\sqrt{\text{VAR}(S)}} \quad \text{if } S < 0. \end{aligned}$$

32.4 Sen's Slope Estimator Test

Sen's estimate is a nonparametric approach for determining the size of a trend in a time series (Sen 1968). Sen's nonparametric method is used to determine the true slope of an existing trend, such as the amount of change every year. A positive Sen's slope value shows an upward or growing trend in the time series, whereas a negative value suggests a downward or decreasing trend.

The Sen's method can be used in cases where the trend can be assumed to be linear that is

$$f(t) = Qt + B \quad (32.6)$$

where Q is the slope, B is the constant, and t is the time.

To get the slope estimate Q , the slopes of all the data value pairs are calculated using the equation:

$$Q_i = \frac{x_j - x_i}{j - i} \quad (32.7)$$

where x_j and x_i are the data values at time j and i ($j > i$), respectively, $i = 1, 2, 3, \dots, N$.

The Sen's estimator of slope is the median of all the values of Q_i .

$$Q = Q_{\left[\frac{(N+1)}{2}\right]}, \text{ if } N \text{ is odd, or}$$

$$Q = \frac{1}{2} \left(Q_{\left[\frac{N}{2}\right]} + Q_{\left[\frac{(N+2)}{2}\right]} \right), \text{ if } N \text{ is even}$$

To obtain an estimate of B in equation $f(t)$ the values of difference $x_i - Qt_i$ are calculated. The median of these values gives an estimate of B .

32.5 Results and Discussion

32.5.1 Statistical Analysis

The arithmetic mean was used to compute the average annual and seasonal rainfall in Mumbai and Pune. The mean annual rainfall in Mumbai is more than in Pune since it is closer to the shore and is in the windward direction, or on the side facing the wind. It is lower in Pune because it is located in the leeward direction, which is shielded from the wind. The average seasonal rainfall in these cities are shown in Figs. 32.1 and 32.2. Both Mumbai and Pune have a similar rainfall pattern, with the majority of

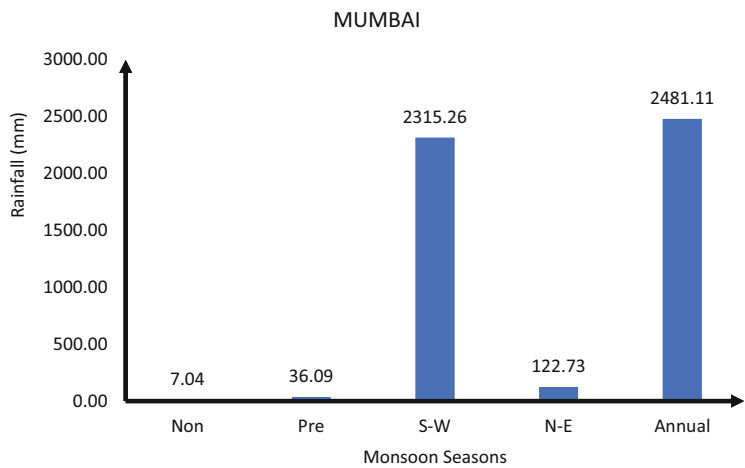


Fig. 32.1 Mean rainfall data of Mumbai

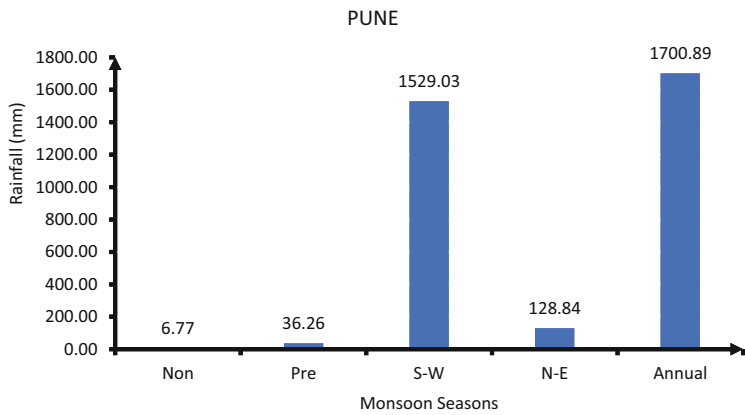


Fig. 32.2 Mean annual rainfall data of Pune

the yearly rainfall falling during the southwest monsoon season and the least falling during the non-monsoon season from January to March.

In Mumbai, where the mean annual rainfall is considerably high, the southwest monsoon rainfall becomes the most significant, accounting for 93.3 percent of the annual rainfall. Pre-monsoon and northeast monsoon both contribute 1.4% and 4.9%, respectively. The non-monsoon contributes the least, on average, 0.28%.

The analysis of mean rainfall distribution reveals that the southwest monsoon rainfall contributes the most, accounting for 89.9% of total annual rainfall whereas, the pre-monsoon rainfall and the northeast monsoon rainfall provide 2% and 7.5%, respectively. Similar to Mumbai, the smallest contribution is from non-monsoon rainfall, which accounts for only 0.3%. Both places have a similar tendency, with

Table 32.1 Statistical analysis of rainfall in Mumbai

Period	Mean (mm)	Maximum (mm)	Minimum (mm)	Standard Deviation	Coeff of Variation	Skewness	Kurtosis
Non	7.04	52.89	0	11.91	1.69	3.24	11.70
Pre	36.09	271.16	0.43	68.63	1.90	2.78	7.46
S-W	2315.26	3133.26	1295.19	564.82	0.24	-0.22	-1.18
N-E	122.73	260.91	12.72	87.05	0.71	0.67	-0.45
Annual	2481.11	3387.31	1326.58	587.23	0.24	0.11	-0.91

Table 32.2 Statistical analysis of rainfall in Pune

Period	Mean (mm)	Maximum (mm)	Minimum (mm)	Standard Deviation	Coeff of Variation	Skewness	Kurtosis
Non	6.77	41.23	0	9.33	1.38	2.71	9.26
Pre	36.26	166.17	2.56	46.49	1.28	1.95	3.06
S-W	1529.03	2462.62	823.02	476.95	0.31	0.21	-0.95
N-E	128.84	286.71	18.92	85.06	0.66	0.63	-0.72
Annual	1700.89	2729.43	890.96	509.23	0.30	0.29	-0.82

greatest rainfall occurring during the southwest and northeast monsoon seasons, followed by decreased rainfall during the pre- and non-monsoon seasons.

The statistical parameters maximum (Max) and minimum (Min) rainfall, mean, standard deviation, skewness, and Kurtosis of rainfall data for both Mumbai and Pune have been computed and are given in Tables 32.1 and 32.2 for seasonal monsoon and annual rainfall data.

The highest mean rainfall value is observed in the S-W monsoon season of 2315.26 mm. In the non-monsoon season, the greatest rainfall was 52.89 mm in 2015, while the smallest rainfall was 0 mm in 2012. The greatest rainfall in the pre-monsoon season was 271.16 mm in 2000, while the smallest rainfall was 0.43 mm in 2013 and the average annual rainfall over the years 2000–2020 in Mumbai is observed to be 2481.11 mm. The maximum rainfall overall annually of 3387.31 mm was recorded in 2019, and minimum annual rainfall of 1326.58 mm was recorded in the year 2002.

The values of skewness are positive in the non-monsoon, pre-monsoon, northeast monsoon, and annual seasons, indicating that it is positively skewed, i.e. the number of observations fewer than the center point (or mean) is greater than the number of values after the center point. In contrast, during the southwest monsoon season, the skewness score is negative, indicating that the data is negatively skewed.

Very high Kurtosis values obtained in non-monsoon and pre-monsoon periods suggest that the data set has a Leptokurtic distribution (with a high degree of peakedness), with a very prominent peak and heavy tails, indicating a dramatic contrast between the peak value and the other values. Negative Kurtosis values in the southwest monsoon, northeast monsoon, and yearly period imply a Platykurtic distribution for the data set (having a low degree of peakedness). They have a flat top around the mean rather than a high peak, implying that the gap between the peak value and the other values in the time period is less.

The highest mean rainfall value is observed in the S-W monsoon season of 1529.03 mm. The maximum rainfall in the non-monsoon season of 41.23 mm was recorded in 2015, and the minimum rainfall of 0 mm in 2012. The maximum rainfall in pre-monsoon season of 166.17 mm was recorded in 2006, and the minimum rainfall of 2.56 mm in 2013. The maximum rainfall overall annually of 2729.43 mm was recorded in 2019, and minimum annual rainfall of 890.96 mm in 2002.

The values of skewness are positive in all the rainfall periods which indicates that they are positively skewed. Very high Kurtosis values in non- and pre-monsoon periods as observed in Mumbai indicate that the data set is Leptokurtic distribution, they have a very distinct peak and the tails are heavy, i.e. sharp difference between the peak value and the other values. Whereas, negative values of Kurtosis in S-W monsoon season, N-E monsoon season, and annual period indicate that the data set is a Platykurtic distribution, they have a flat top around the mean instead of a sharp peak so there is less difference between the peak value and the other values. This exact same trend was observed in Mumbai also.

The percentage of coefficient of variation of rainfall in Mumbai as shown in Fig. 32.3. The S-W monsoon season contributes the most to yearly rainfall, the numbers are closer together, resulting in a comparable coefficient of variation, 24.40% for S-W and 23.67% for the annual period. In contrast, there is a wide range of values for pre-monsoon (190.15%), non-monsoon (169.25%), and northeast monsoon (70.93%) (Fig. 32.4).

Similar to Mumbai, the S-W monsoon period contributes the bulk of yearly rainfall in Pune, and the values are closer to each other, resulting in a similar coefficient of variation, 31.19% for the S-W monsoon season and 29.94% for the annual period. Higher values are observed during the non-monsoon season (137.84%), pre-monsoon season (128.21%), and N-E monsoon season (66.02%).

The graphs and the numbers generated in the table show that when the coefficient of variation is low, the skewness is also low, as both represent how the distribution is with regard to the mean value and provide insight into the pattern of data values over time. As a result, the coefficient of variation and skewness values are both low for the S-W monsoon season and the yearly period, whereas they are high for the non-, pre-,

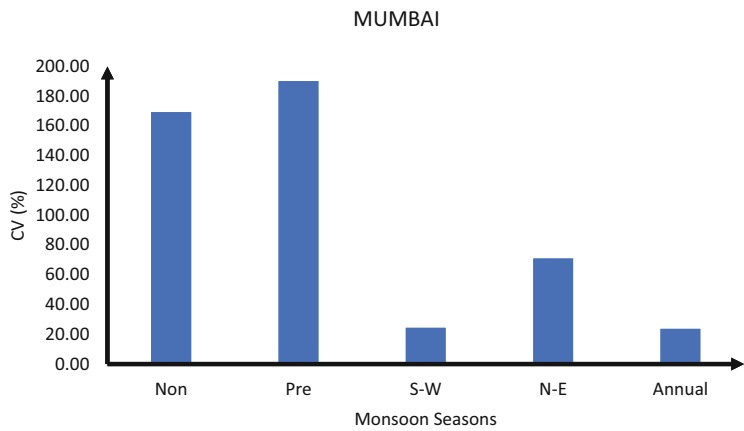


Fig. 32.3 Coefficient of variation (%) of rainfall in Mumbai

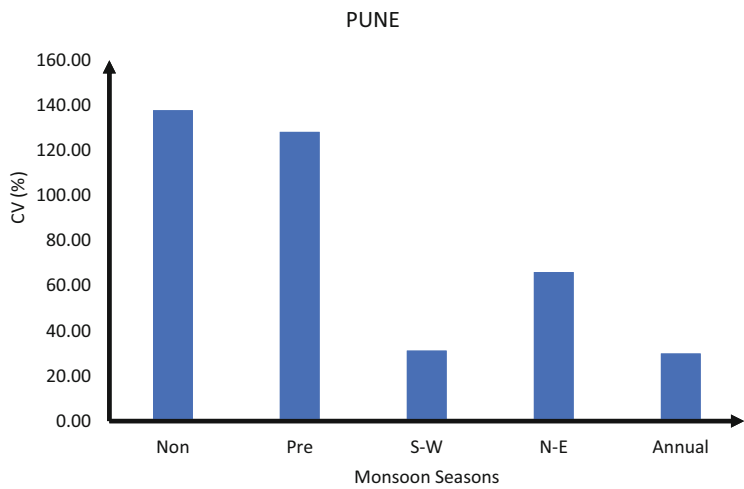


Fig. 32.4 Coefficient of variation (%) of rainfall in Pune

and N-E monsoon seasons. The standard deviation of seasonal and annual rainfall in both cities are shown in Fig. 32.5.

32.5.2 Rainfall Trend Analysis

The rainfall trends from the year 2000–2020 for Mumbai and Pune using monthly rainfall data have been analyzed by Mann-Kendall, Sen’s Slope Estimator, and Linear Regression Analysis. Figures 32.6, 32.7, 32.8, 32.9, and 32.10 show the

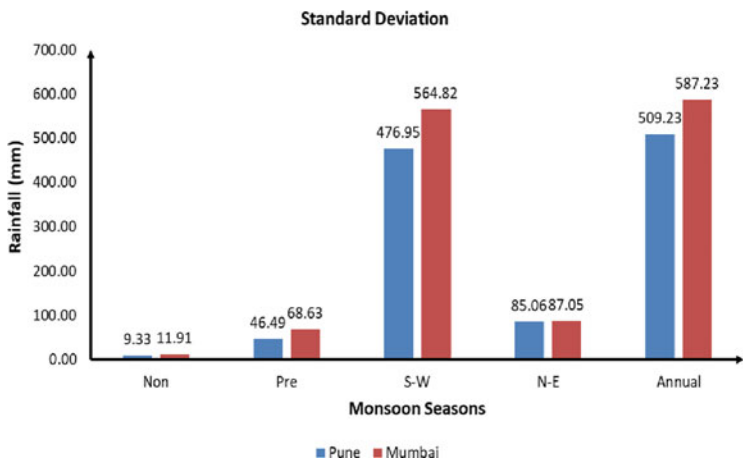


Fig. 32.5 Standard deviation of annual and seasonal rainfall in Pune and Mumbai

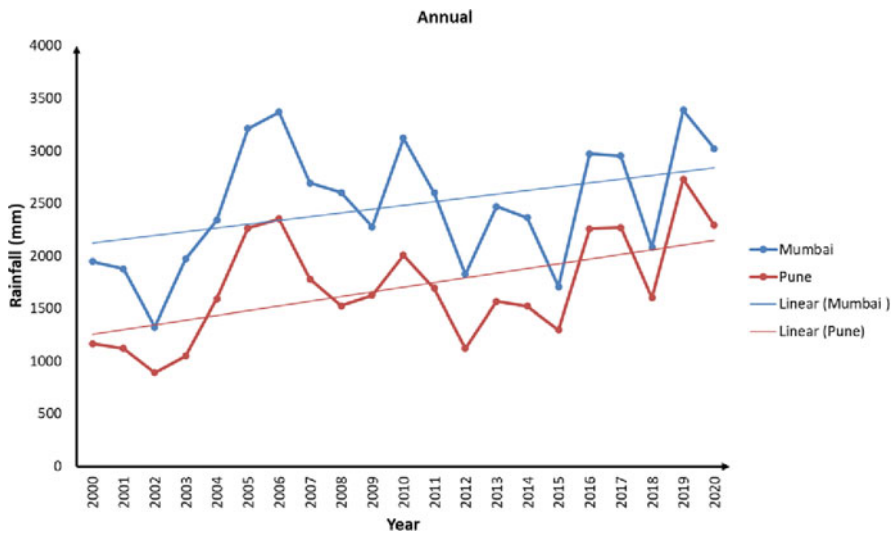


Fig. 32.6 Annual rainfall trend in Mumbai and Pune

trend of rainfall for annual, non-monsoon, pre-monsoon, S-W monsoon, and N-E monsoon over the period 2000–2020. In both Mumbai and Pune, the total annual rainfall rise. This might be attributable to a number of factors, including climate change, which raises overall temperatures and increases precipitation. During the non-monsoon season in the provided time period, a rising tendency has been noted. During the pre-monsoon season (April–May), both Mumbai and Pune had a declining tendency, but the southwest monsoon period in both cities saw an increase.

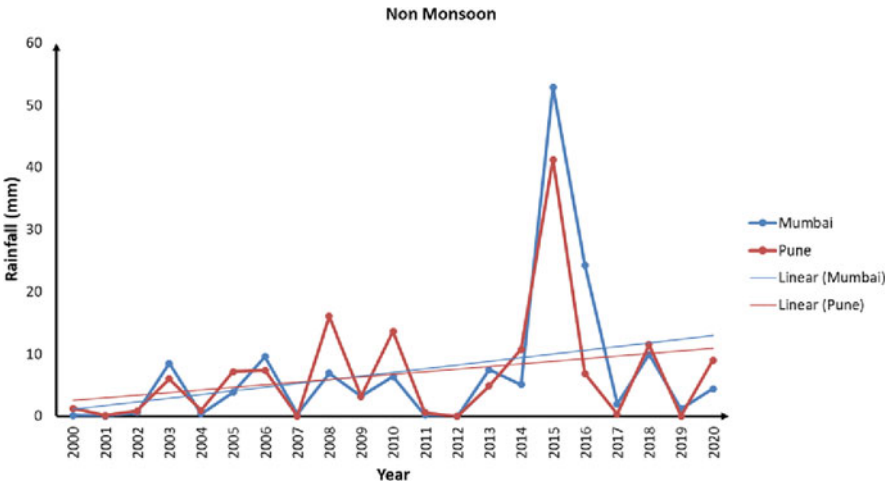


Fig. 32.7 Non-monsoon rainfall trend in Mumbai and Pune

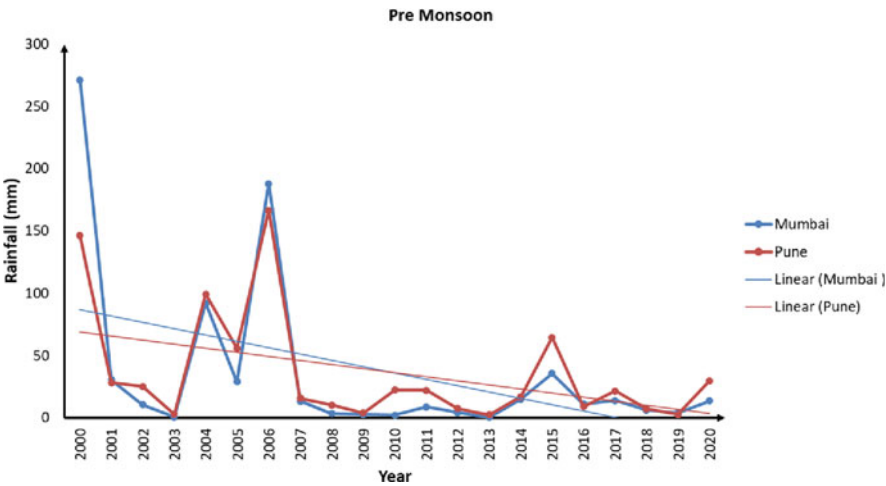


Fig. 32.8 Pre-monsoon rainfall trend in Mumbai and Pune

During the northeast monsoon season, a rising tendency has been noted in both Mumbai and Pune.

32.5.3 Mann-Kendall Analysis

This test just identifies the direction of the trend, not the amount of the trend, therefore it does not need presuming normality and is entirely dependent on the

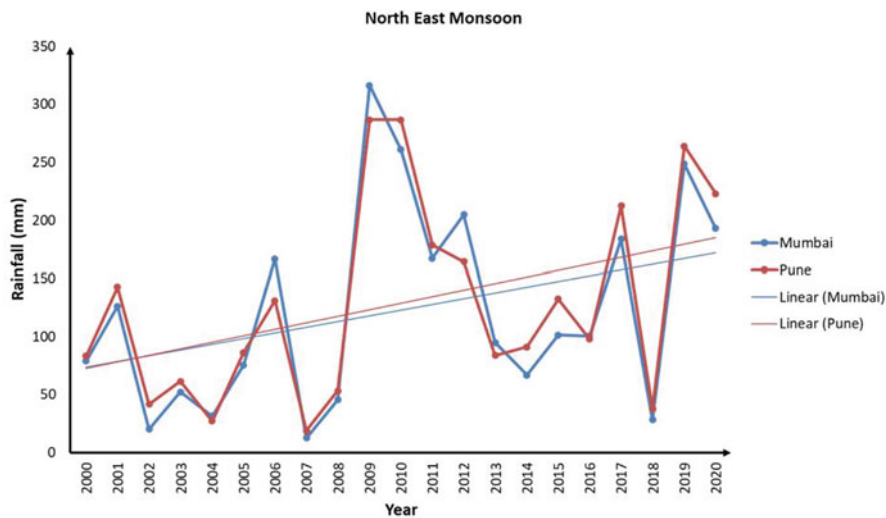


Fig. 32.9 Southwest monsoon rainfall trend in Mumbai and Pune

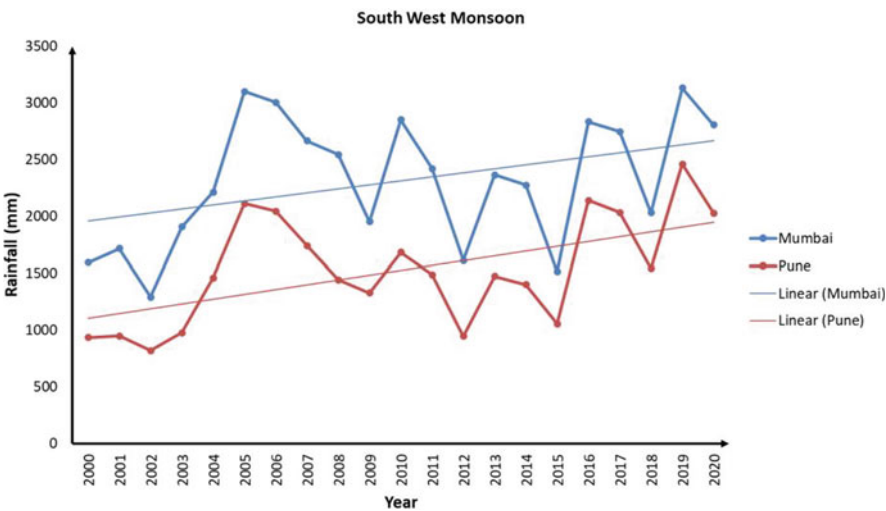


Fig. 32.10 Northeast monsoon rainfall trend in Mumbai and Pune

numbers that have been recorded and utilized for the computation. If the Z value is negative, the trend is said to be falling; if the Z value is positive, the trend is said to be advancing. The results of Mann-Kendall analysis are given in Table 32.3

The results suggest that rainfall in Mumbai and Pune is increasing in the non-monsoon season, S-W monsoon season, N-E monsoon season, and yearly period. During the pre-monsoon season, it has been declining. Mann-Kendall analysis of seasonal and annual rainfall of Mumbai and Pune are shown in Fig. 32.11.

Table 32.3 Mann-Kendall—Z values

Area/Season	Annual	Non	Pre	S-W	N-E
Mumbai	1.358861	1.479649	-1.17768	1.479649	1.600437
Pune	2.204375	0.875711	-1.84201	2.204375	1.902406

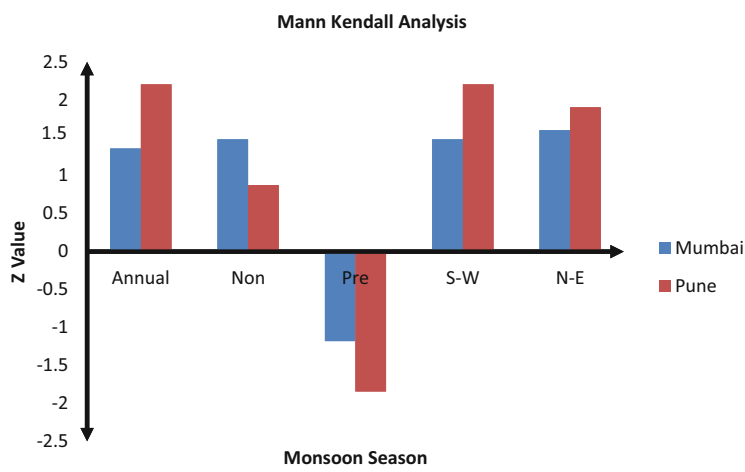


Fig. 32.11 Mann-Kendall analysis of annual and seasonal rainfall

32.5.4 Sen’s Slope Estimator

Sen's Slope estimator findings are given in Table 32.4. Sen’s Slope method analysis for annual, non-monsoon season, pre-monsoon season, S-W monsoon season, N-E monsoon season indicates rising patterns in annual, non-, S-W, N-E monsoon seasons, and declining trends in pre-monsoon season, comparable to Mann-Kendall Analysis. The S-W monsoon season has the most yearly growth in Mumbai, with 39.44 mm and 40.08 mm in Pune, followed by the N-E monsoon season’s 4.18 mm/year in Mumbai and 0.28 mm/year in Pune, and the non-monsoon season’s 0.21 mm/year rise. However, there has been a 0.8 4 mm/year reduction in Mumbai and a 1.40 mm/year decline in Pune during the pre-monsoon season. On an annual basis, Mumbai has had a rise of 41.31 mm/year, while Pune has seen a rise of 46.21 mm/year.

Table 32.4 Sen’s slope estimator

Rainfall Season	Mumbai		Pune	
	Slope	Constant	Slope	Constant
Annual	41.31353	2102.728	46.20932	1137.668
Non	0.214917	1.050833	0.280769	0.969231
Pre	-0.84053	22.22	-1.40403	34.06653
S-W	39.44528	1948.687	40.08409	1009.871
N-E	4.185417	50.3275	5.762083	51.7175

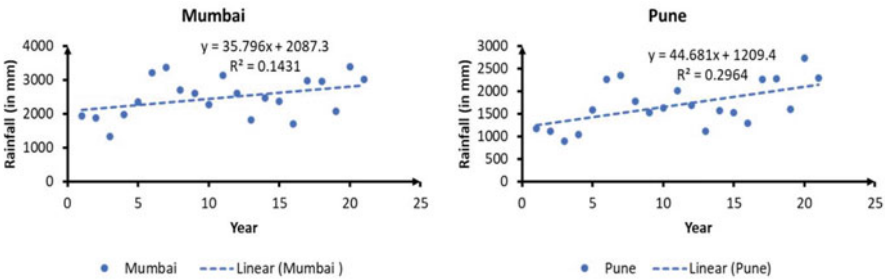


Fig. 32.12 Linear regression analysis of annual rainfall

32.5.5 Linear Regression Analysis (Figs. 32.12, 32.13, 32.14, 32.15, and 32.16)

The slope and intercept values of the linear regression analysis graphs are given in Tables 32.5 and 32.6. The slope values in Mumbai and Pune have a negative sign in the pre-monsoon season, indicating that there has been an overall declining tendency throughout the years. Similarly, slope values in Mumbai and Pune have a positive sign in the non-monsoon, S-W, N-E monsoon seasons, and the yearly period, showing an overall growing tendency (Fig. 32.17).

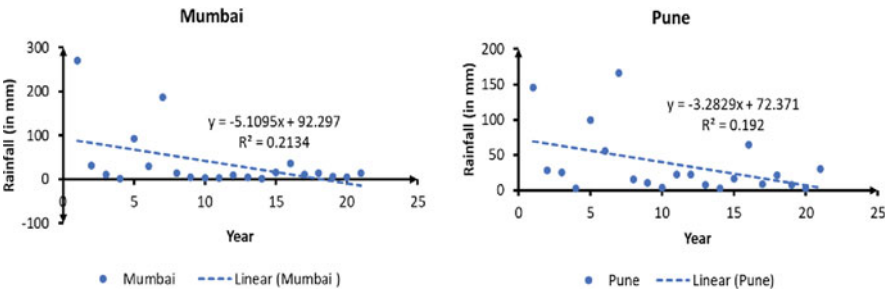


Fig. 32.13 Linear regression analysis of non-monsoon season

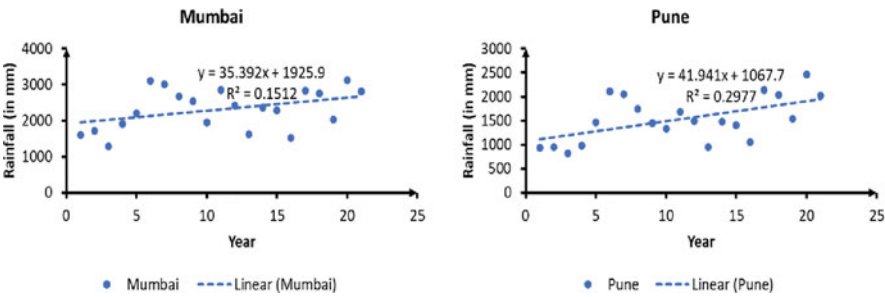


Fig. 32.14 Linear regression analysis of pre-monsoon season

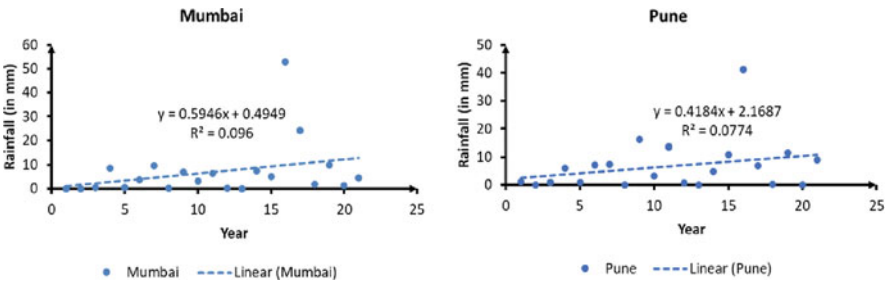


Fig. 32.15 Linear regression analysis of S-W monsoon season

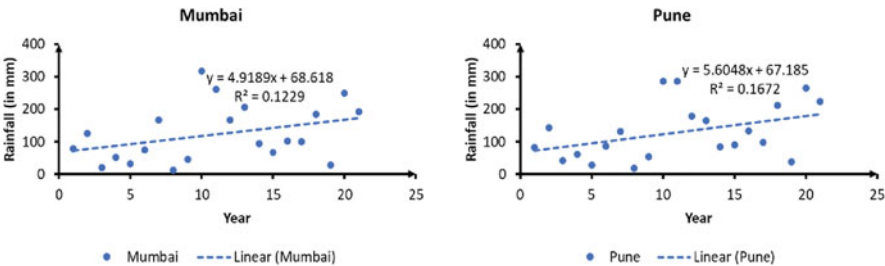


Fig. 32.16 Linear regression analysis of N-E monsoon season

Table 32.5 Linear regression analysis for annual and seasonal rainfall

Rainfall Season	Mumbai		Pune	
	Slope	Intercept	Slope	Inercept
Annual	35.8	2087	44.68	1209
Non	0.59	0.49	0.42	2.17
Pre	-5.11	92.3	-3.28	72.37
S-W	35.39	1926	41.94	1068
N-E	4.919	68.62	5.605	67.18

Table 32.6 Comparison between Sen slope and regression analysis

Slope/Monsoon Season	Annual	Non	Pre	S-W	N-E
Mumbai					
Sen Slope	41.31353	0.214917	-0.84053	39.44528	4.185417
Regression Slope	35.8	0.59	-5.11	35.39	4.919
Pune					
Sen Slope	46.20932	0.280769	-1.40403	40.08409	5.762083
Regression Slope	44.68	0.42	-3.28	41.94	5.605

32.5.6 Reliability Analysis

Internal consistency means the amount to which test measurements stay consistent across repeated trials under the same conditions is referred to as dependability. The design of drinking water schemes is based on 90% dependability, whereas irrigation requirements are calculated using 75% dependability. The results are shown in Figs. 32.18, 32.19, 32.20, 32.21, and 32.22. The results showed that the rainfall during non-monsoon and pre-monsoon period would not be having any significant contribution in the annual rainfall. S-W monsoon has significant contribution in annual rainfall.

90% dependable annual rainfall in Mumbai and Pune is 1824.6 mm and 1121 mm, 75% dependable annual rainfall in Mumbai and Pune is around 1970 mm and 1295 mm. During non-monsoon period 90% dependable rainfall for Mumbai and Pune is 0.12 mm and 0.03 mm. While 75% dependable rainfall is approximately 0.4 mm for Mumbai and 0.62 mm for Pune. In pre-monsoon season 90% dependable

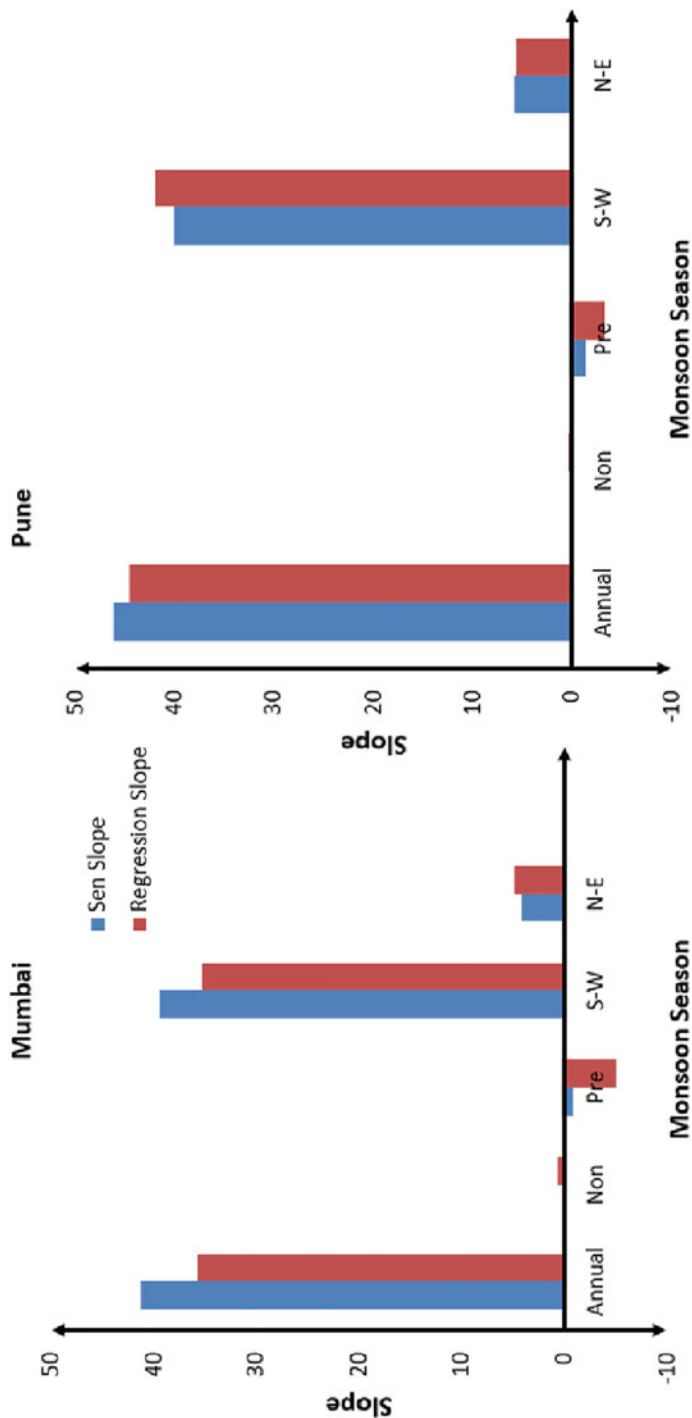


Fig. 32.17 Comparison of Sen slope and regression analysis

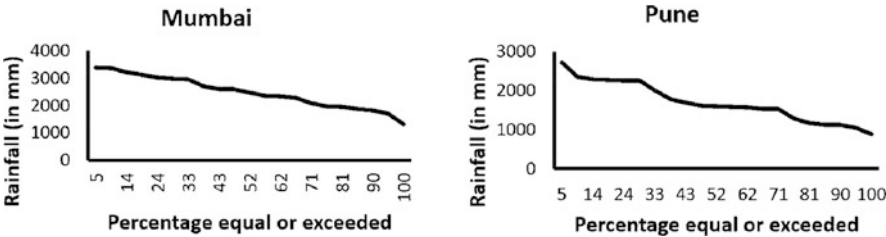


Fig. 32.18 Reliability curve for annual rainfall

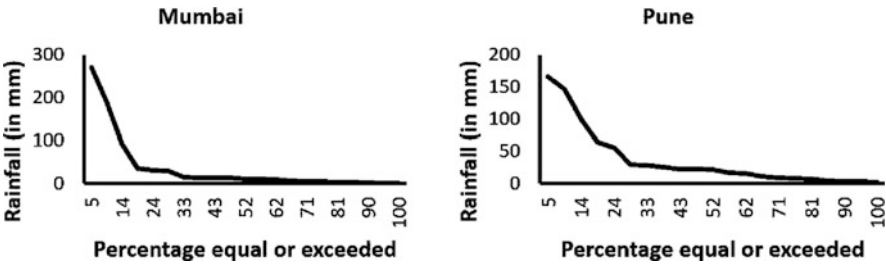


Fig. 32.19 Reliability curve for non-monsoon season

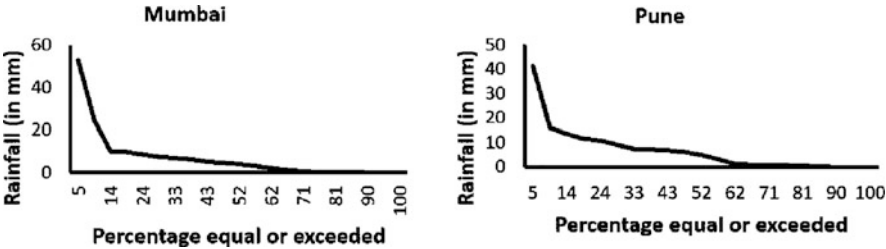


Fig. 32.20 Reliability curve for pre-monsoon season

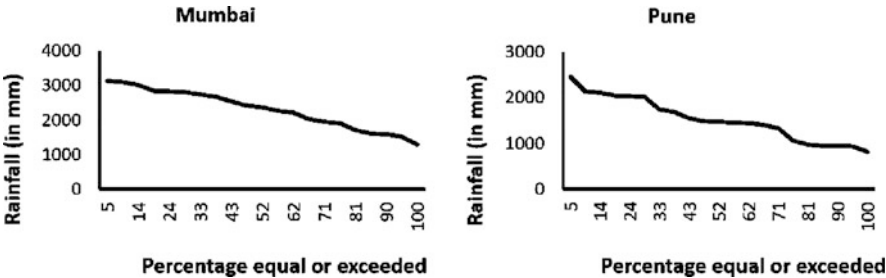


Fig. 32.21 Reliability curve for southwest monsoon season

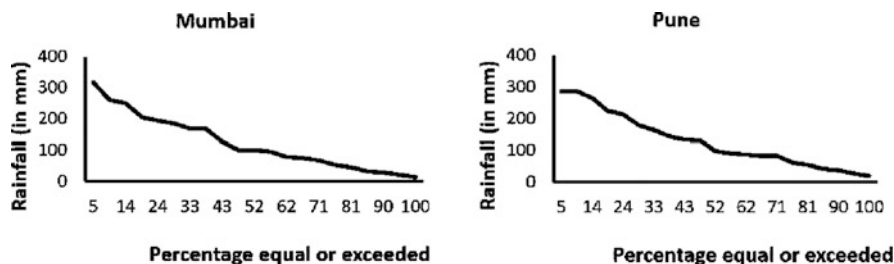


Fig. 32.22 Reliability curve for northeast monsoon season

rainfall for Mumbai and Pune are 2.2 mm and 2.9 mm and 75% dependable rainfall is around 4.2 mm and 7.5 mm for Mumbai and Pune.

During S-W monsoon the 90% dependable rainfall in Mumbai and Pune is 1594.4 mm and 947.9 mm. In northeast monsoon season the 90% dependable rainfall for Mumbai is 28.4 mm and for Pune around 36.7 mm.

32.6 Conclusion

Monthly rainfall data of Mumbai and Pune cities from 2000 to 2020 were analyzed in this study. Annual and seasonal rainfall are statistically analyzed. The outcomes are as follows: In Mumbai and Pune, the average annual rainfall is 2481.11 mm and 1700.89 mm, respectively. The southwest monsoon has contributed the most to both cities, accounting for 93.3% in Mumbai and 89.9 percent in Pune. In both Mumbai and Pune, a rising tendency has been noted in the annual period, non-monsoon, southwest monsoon, and northeast monsoon. In both locations, however, a diminishing tendency has been noted throughout the pre-monsoon season.

The highest rise per year in Mumbai is 39.44 mm/year in the S-W monsoon season, 4.18 mm/year in the N-E monsoon season, and 0.21 mm/year in the non-monsoon season. During the pre-monsoon season, there was a 0.84 mm/year reduction. A rise of 41.31 mm/year has been noted on a yearly basis.

The annual rise in Pune is 40.08 mm/year in the S-W monsoon season, followed by a 5.76 mm/year increase in the N-E monsoon season and a 0.28 mm/year increase in the non-monsoon season. During the pre-monsoon season, there was a 1.40 mm/year decline. A rise of 46.21 mm/year has been noted on a yearly basis.

The dependable rainfall (90% or 75%) varies widely between the months and it is the highest in the southwest monsoon season. These values slightly decrease in the following months during northeast monsoon season. Thereafter, the rainfall reduces drastically and 90% and 75% rainfall are almost nil for non-monsoon and pre-monsoon season. Thus, this shows the need for very careful planning in the design of water related schemes for drinking, irrigation or industry. This also highlights the need for creation of storage to meet the demand in these months.

References

- Burn DH, Elnur MAH (2002) Detection of hydrologic trends and variability. *J Hydrol* 255(1–4): 107–122
- Gandhre N, Dauji S, Londhe S (2020) Analysis of precipitation extremes for coastal districts of Maharashtra. In: Proceedings of the international conference on recent advances in computational techniques (IC-RACT)
- Jagadeesh P, Agrawal S (2015) Investigation of trends and its magnitude by non-parametric Mann-Kendall and Sen's slope methods. *Int J Hydrol Sci Technol* 5(1):83–94
- John CK, Pu JH, Pandey M, Hanmaiahgari PR (2021a) Sediment deposition within rainwater: case study comparison of four different sites in Ikorodu, Nigeria. *Fluids* 6(3):124
- John CK, Pu JH, Moruzzi R, Pandey M (2021b) Health-risk assessment for roof-harvested rainwater via QMRA in Ikorodu area, Lagos, Nigeria. *J Water Clim Change* 12(6):2479–2494
- Joshi MK, Pandey AC (2011) Trend and spectral analysis of rainfall over India during 1901–2000. *J Geophys Res Atmos* 116:D6
- Kendall MG (1975) Rank correlation methods. Google Scholar, London
- Kumar V, Jain SK, Singh Y (2010) Analysis of long-term rainfall trends in India. *Hydrol Sci J* 55(4):484–496
- Mann HB (1945) Nonparametric tests against trend. *Econometrica* 13:245–259
- Mondal A, Khare D, Kundu S (2015) Spatial and temporal analysis of rainfall and temperature trend of India. *Theor Appl Climatol* 122(1):143–158
- Mooley DA, Parthasarathy B (1984) Fluctuations in all-India summer monsoon rainfall during 1871–1978. *Clim Chang* 6(3):287–301
- Panda A, Sahu N (2019) Trend analysis of seasonal rainfall and temperature pattern in Kalahandi, Bolangir and Koraput districts of Odisha, India. *Atmos Sci Lett* 20(10):e932
- Pandey M, Pu JH, Pourshahbaz H, Khan MA (2022) Reduction of scour around circular piers using collars. *J Flood Risk Manage* 3:e12812. <https://doi.org/10.1111/jfr3.12812>
- Parthasarathy B, Munot AA, Kothawale DR (1994) All-India monthly and seasonal rainfall series: 1871–1993. *Theor Appl Climatol* 49(4):217–224
- Patil JP, Sarangi A, Singh DK, Chakraborty D, Rao MS, Dahiya S (2013) Rainfall trend analysis: a case study of Pune district in Western Maharashtra region. *J Soil Water Conserv* 12(1):35–43
- Potdar SS, Kulkarni S, Patil P, Pawar RP, Jakhalekar VV, Nade DP (2019) The long-term trend analysis of rainfall data from 1901 to 2015 for Maharashtra and Goa region from India. *Int J Water* 13(3):293–309
- Pu JH, Wallwork JT, Khan M, Pandey M, Pourshahbaz H, Satyanaga A, Hanmaiahgari PR, Gough T (2021) Flood suspended sediment transport: combined modelling from dilute to hyper-concentrated flow. *Watermark* 13(3):379
- Rana A, Uvo CB, Bengtsson L, Parth Sarthi P (2012) Trend analysis for rainfall in Delhi and Mumbai, India. *Clim Dyn* 38(1):45–56
- Rana N, Singh S, Sundriyal YP, Juyal N (2013) Recent and past floods in the Alaknanda valley: causes and consequences. *Curr Sci* 105(9):1209–1212
- Sen PK (1968) Estimates of the regression coefficient based on Kendall's tau. *J Am Stat Assoc* 63(324):1379–1389
- Shankar MS, Pandey M, Shukla AK (2021) Analysis of existing equations for calculating the settling velocity. *Watermark* 13(14):1987
- Singh UK, Jamei M, Karbasi M, Malik A, Pandey M (2022) Application of a modern multi-level ensemble approach for the estimation of critical shear stress in cohesive sediment mixture. *J Hydrol* 607:127549
- Wallwork JT, Pu JH, Kundu S, Hanmaiahgari PR, Pandey M, Satyanaga A, Pandey M, Satyanaga A, Amir Khan M, Wood A (2022) Review of suspended sediment transport mathematical modelling studies. *Fluids* 7(1):23